ArogyaSaathi: project ko mentor ko kaise samjhayein

Voice-First Triage, Chronic Care aur Outbreak Early Warning | Tech for Impact | Deviathon 2.0

30 second ka pitch (yeh pehle bolna)

*"Gaon mein doctor door hota hai aur ASHA didi ke paas decision lene ke liye koi support nahi hota. ArogyaSaathi ek phone app hai jisme ASHA mareez ki baat apni bhasha mein bol ke bata sakti hai, aur app turant batata hai: **Emergency, Refer, Treat locally ya Home care**. Khatre ke lakshan (jaise seene mein dard + pasina) pakadne ka kaam **fixed rules** karte hain, AI nahi, isliye koi khatra miss nahi hota. Saath mein BP-sugar ka follow-up chalta hai, aur gaon ke clinic, dawai ki dukaan aur school ki absence ke numbers milakar **outbreak ka alert** pehle hi mil jata hai. Har decision ka audit trail rehta hai aur doctor ka human oversight hamesha rehta hai."*

Problem jo hum solve karte hain (DEV_107)

- **Deterioration miss hota hai:** BP/sugar ke mareez sirf kabhi-kabhi doctor ko dikhte hain, beech mein badhta sugar + chhooti dawai kisi ko dikhti nahi.
- **ASHA ko decision support nahi:** treat karein, refer karein ya emergency? Yeh unsure rehta hai.
- **Outbreak late pakda jata hai:** clinic, pharmacy aur school ke signals kabhi ek saath dekhe nahi jaate.
- **Bhasha ki dikkat:** mareez English ya text-based system use nahi kar pate.
- **AI ka safety risk:** AI andaza laga ke dawai bata de ya red flag miss kar de to nuksaan ho sakta hai. Isliye guardrails + audit zaroori.

Hamare 4 users aur har ek ko kya milta hai

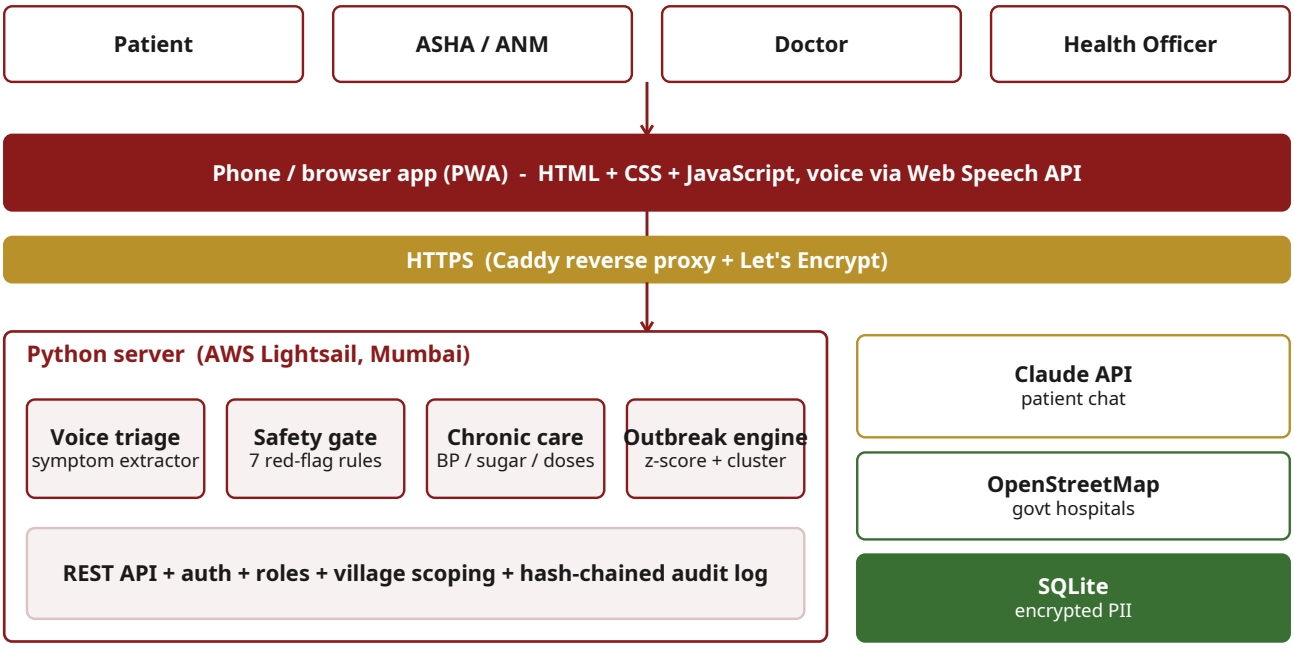
User	App mein kya karta hai
Patient (mareez)	Apni BP/sugar dekhta hai, dawai li ya nahi mark karta hai, doctor ke notes aur hospital referral padhta/sunta hai, 5 bhasha mein AI assistant se poochta hai, 1 tap mein 108 call.
ASHA / ANM	Mareez register karti hai (6 digit activation code deti hai), voice triage karti hai, BP/sugar reading daalti hai, nearest sarkari hospital ko refer karti hai, roz ki village report (chemist + school counts) bhejti hai.
Doctor (PHC)	Escalation queue dekhta hai (Emergency pehle), one-page summary padhta hai, AGREE ya OVERRIDE karta hai, mareez ko note bhejta hai. Uska feedback labelled data ban jata hai.
Health Officer	Village map par outbreak alerts, syndromic signals ke charts, alert acknowledge karna, audit trail.

1. Tech stack ek nazar mein (jo sach mein use hua)

Layer	Technology	Kyun chuna
Frontend	Plain HTML + CSS + JavaScript (koi framework nahi). Installable PWA: manifest.json + service worker.	Halka, saste phone par bhi tez; Play Store ke bina "Install app" ban jata hai; offline app shell.
Voice input	Browser ki Web Speech API (SpeechRecognition) - hi-IN, en-IN, bn-IN, mr-IN, te-IN. Text box hamesha editable.	Server par STT model chalane ka kharcha nahi; ASHA bol ke ya type karke dono kar sakti hai.
Voice output	speechSynthesis (browser) - AI jawab / notes ko padh ke sunata hai.	Low-literacy mareez ke liye.
Map	Leaflet 1.9.4 + OpenStreetMap tiles.	Free, no API key; village outbreak map.
Backend	Python 3.12, standard-library http.server (ThreadingHTTPServer), REST JSON API (/api/v1).	Sirf 2 pip libraries lagti hain; deploy aur samajhna aasan.
Database	SQLite (ek file), hash-chained audit_log, 14 tables.	Hackathon scale par zero-setup; backup sirf file copy.
AI (chat)	Anthropic Claude API, model claude-opus-5-5, effort "low", server-side fallback beta. Python SDK anthropic.	Mareez ke sawalon ka jawab 5 bhasha mein; strict system prompt (no diagnosis, no medicine).
Triage engine	Hamari apni Python state machine (langgraph_triage.py) + symptom extractor (dictionary + negation).	Predictable aur auditable; har rule ka code (RF-...) log hota hai.
Security	scrypt password hash, Fernet (cryptography lib) se naam/mobile encrypt, random session tokens (SHA-256 hashed), pseudonym IDs.	DPDP-style privacy: database leak ho to bhi naam/number nahi padhe jaate.
Hospital finder	OpenStreetMap Overpass API, fallback Nominatim; haversine distance; name-regex se govt vs private chhaanta.	Free; sarkari PHC/CHC/district hospital dhoondne ke liye.
Deploy	AWS Lightsail (Mumbai), Ubuntu 24.04, systemd service, Caddy auto-HTTPS (Let's Encrypt), domain 16-4-93-27.sslip.io, daily DB backup (cron 02:30, 14 din).	Phone par HTTPS zaroori (mic, GPS aur PWA install ke liye).

Server ki dependencies sirf do hain: `anthropic==2.3.0` aur `cryptography==50.0.2`. Baaki sab Python standard library hai.

2. Architecture



Safety gate sits BEFORE the AI: a red flag never reaches the model.

Flow ek line mein: ASHA bolti hai -> browser text banata hai -> server symptoms nikalta hai -> safety gate check karta hai -> triage level + checklist wapas -> agar Emergency/Refer hai to doctor ki queue mein case + one-page summary -> sab kuch audit log mein.

3. Har feature kaise kaam karta hai

A. Voice triage (ASHA ka main kaam)

- ASHA consent leti hai (voice consent ke bina triage start hi nahi hota), mareez chunti hai ya walk-in ki umar/gender daalti hai, aur bhasha chunti hai.
- **Symptom extractor** (symptom_extractor.py): Hindi/English/Bengali/Marathi/Telugu ke shabdon ki dictionary se symptoms ke standard codes banata hai (jaise *chest_pain, sweating, radiating_pain*). "Bukhar nahi hai" jaisi **negation** samajhta hai, aur din/number words bhi.
- **Guided questions:** agar jaankari kam ho to app agla sawal poochta hai (seene ka dard baayein haath mein ja raha hai? bachcha doodh pi pa raha hai?). MVP mein 3 conditions: **fever, child danger signs (IMNCI), chest pain**.
- Output: **Emergency / Refer / Treat locally / Home care** + step-by-step checklist + protocol citation + uncertainty flag.

B. Rule-based safety gate (sabse important baat)

safety_gate.py mein **7 fixed rules** hain. Inme se koi match ho to triage turant **EMERGENCY** hota hai aur AI/model ko bulaya hi nahi jata.

Rule code	Kab trigger hota hai
RF-CHEST-PAIN-01	Chest pain + (pasina / haath-jabde mein dard / saans phoolna / chakkar)
RF-PEDIATRIC-DANGER-01	5 saal se chhota bachcha + (jhatke / doodh-paani nahi pi raha / sab ulti / bahut sust)
RF-RESPIRATORY-01	Severe saans ki dikkat, stridor, pasli andar dhansna, neele hont
RF-NEURO-FEVER-01	Bukhar + (gardan akdi / hosh mein badlav / rash) - meningitis ka shaq
RF-MATERNAL-01	Garbhavati mein bahut bleeding / tez sir dard + dhundhla dikhna / samay se pehle dard
RF-VITALS-HYPOXIA	SpO2 < 90%
RF-VITALS-HYPERTENSIVE-CRISIS	Upper BP >= 180 mmHg

- **Zero prescription:** `sanitize_zero_prescriptions()` output mein dawai ke naam/doses (jaise amoxicillin, metformin, "take 500mg") dhoondh ke rok deta hai. Referral summary mein `zero_prescription_guaranteed = True` hota hai.
- Chat mein bhi har message pehle isi gate se guzarta hai. Danger sign mila to **fixed emergency reply** jata hai (5 bhasha mein) aur doctor ko escalation banta hai; Claude ko call hi nahi hoti.

C. Chronic care (BP + sugar + dawai)

chronic_care.py reading history se trend nikalta hai. ASHA reading daalti hai (mareez khud reading nahi daalta, sirf dawai li/nahi li mark karta hai).

Check	Rule
Hypertensive crisis	Upper BP >= 180 ya lower >= 110
Sustained Stage-2 BP	BP >= 140/90 aur pichhli 3 mein se 2 readings >= 140
Accelerating trend	Pichhle half ka average pehle half se >= 8 mmHg zyada
Sugar	< 70 hypoglycemia; >= 250 severe; fasting >= 140 ya >= 200 sustained (2 of last 3 >= 180)
Dawai adherence	Lagatar 3 miss = CRITICAL_MEDICATION_DROPOUT; 1+ miss = chetavni; adherence < 70% = poor

Risk level: **LOW / MODERATE / HIGH / CRITICAL**. Critical par "turant ASHA home visit + doctor tele-consult", High par 24-48 ghante mein. Aisi risk badhne par doctor ki queue mein automatic escalation banta hai.

D. Outbreak early warning (3 signals ek saath)

Teen stream milti hain: (1) clinic ke fever/diarrhoea cases jo voice triage se aate hain, (2) chemist ki ORS aur fever-dawai ki bikri, (3) school mein absent bachche. Last two ASHA roz ki **Daily Report** mein daalti hai.

- **Z-score:** aaj ki value ko pichhle 14 din ke average se compare karte hain: z = (aaj - mean) / (std + 0.5).
- **Fused score:** streams ko weights se milate hain (clinic cases 0.40, ORS sales 0.35, baaki signals 0.25). Level: **WATCH >= 1.5, WARNING >= 2.2, OUTBREAK >= 3.0**.
- **Baseline rule:** scoring tabhi shuru hoti hai jab gaon ke 7+ din ke reports ho chuke hon. Isse naye gaon mein false alarm nahi aata.
- **Corroboration rule (humne jodi):** WARNING/OUTBREAK tabhi jab kam se kam 2 streams z >= 2 par hon, ya excess 10+ cases ka ho. Nahi to level sirf WATCH par rukta hai. Wajah: chhote gaon mein 0 case ke baad achanak 2 case se z-score fool jaata tha.
- **Spatio-temporal clustering:** jo gaon flagged hon aur ek doosre se **10 km** ke andar hon, unhe ek cluster/alert banate hain (haversine distance).
- Officer map par gaon dikhte hain; alert acknowledge hota hai aur audit mein jata hai.

E. Patient AI assistant (Claude)

- Patient 5 bhasha mein type ya bol ke poochta hai; jawab usi bhasha mein aata hai aur padh ke bhi sunaya ja sakta hai.
- Claude ko sirf yeh data milta hai: bhasha, umar, conditions, latest BP/sugar, doctor ke recent notes (naam/number nahi).
- **System prompt ke rules:** diagnose nahi, dawai ka naam/dose nahi, urgent lage to 108, naye problem par ASHA/PHC, 120 shabdon se chhota jawab.
- Error ho ya key na ho to fixed "assistant available nahi, ASHA se milein, emergency mein 108" wala safe jawab.

F. Referral + hospital finder

- ASHA mareez ke gaon ke GPS se **nearest sarkari facility** dhoondti hai: district hospital, CHC, PHC, sub-centre, medical college (OpenStreetMap se; naam ke regex se private hospital hata diye jaate hain; result 24 ghante cache).
- Overpass band ho to Nominatim par fallback; baar-baar fail hone par 1 ghante ke liye circuit breaker.
- Referral mareez ke app mein "Doctor aur Hospital" tab mein dikhta hai (hospital ka naam, level, phone).
- **Doctor ko one-page structured summary** (symptoms, vitals, history, ASHA ke actions, protocol citations) 1 second se kam mein banti hai.

G. Audit aur oversight

Har input, fired rule, model output aur doctor ka override **audit_log** mein jata hai. Har entry mein pichhli entry ka hash judta hai (blockchain jaisa chain): koi purani entry badle to chain toot jaati hai aur `verify_audit_chain()` pakad leta hai. Doctor ka AGREE/OVERRIDE **labelled training data** ke roop mein save hota hai.

4. Security aur privacy

Cheez	Kaise
Password	scrypt (n=2^14, r=8, p=1) + har user ka random salt; hmac.compare_digest se check.
Session	32-byte random token; database mein sirf uska SHA-256 hash; 30 din ki expiry; logout par delete.
PII encryption	Naam aur mobile Fernet (AES) se encrypted; login lookup ke liye alag mobile hash.
Pseudonym ID	Har mareez ka HASH-VILLAGE-xxxx jaisa ID hota hai (audit/logs mein sirf yahi). Doctor ko ilaaj ke liye naam bhi dikhta hai; Health Officer ko sirf gaon-level aggregate numbers.
Consent	Voice consent ke bina na registration, na triage.
Access control	Role-based + village-scoped: ASHA sirf apne gaon ke mareez dekhti hai. Staff sign-up ke liye access code; mareez ko ASHA ka 6-digit activation code (72 ghante valid).
Web	HTTPS, path-traversal check (static files), no-store cache on API, security headers.

5. Mentor ko sach batane wali baatein (isse pakde nahi jaoge)

Mentor code dekh sakta hai. Neeche wali cheezein repo mein dikhti hain, isliye pehle khud bata dena sabse safe hai. Problem statement mein jo stack suggest hua tha aur jo humne sach mein use kiya, dono alag hain.

Pehle se bata do: "Maine scope aisa rakha"

"Problem statement mein FastAPI, PostgreSQL, Celery, LangGraph jaisi cheezein suggest thi. Hackathon mein ek chalne wala, deploy hua prototype dena zaroori tha, isliye humne sabse halka stack liya: Python standard library + SQLite. Architecture modular hai, to scale ke time par Postgres/PostGIS lagana aasan hai."

Sawal jo mentor utha sakta hai	Sach jawab
"Kya yeh LangGraph hai?"	Nahi. File ka naam langgraph_triage.py hai, par yeh hamari khud ki Python state-machine hai (multi-turn conversation state + rules). LangGraph library install nahi hai. Aisa na bolein ki LangGraph use kiya.
"Repo mein FastAPI / Postgres / Celery / Redis dikh raha hai"	backend/app mein api/, crud/, models/, tasks/, main.py aur requirements.txt purana scaffold hai jo problem statement ke stack ke hisaab se shuru mein bana tha. **Chalne wala server ye nahi hai.** Asli server local_server.py (http.server) + portal/ + SQLite hai, aur server ki dependencies requirements-server.txt mein sirf 2 hain. Scaffold ko "future scale path" bol sakte hain.
"Speech-to-text kaun sa model hai?"	Browser ka Web Speech API (Chrome mein Google ki speech service; internet chahiye). Bhashini/IndicConformer ka stt_service.py repo mein hai par abhi jura nahi hai. Offline voice abhi nahi hai.
"Symptom extraction ML hai?"	Nahi, dictionary + negation + number rules hain. Kyunki rules deterministic hain, safety ke liye acchha hai; par naye words ke liye dictionary badhani padegi. Target "F1 >= 0.85" abhi formal test set par naapa nahi gaya.
"AI chat chal raha hai?"	Code taiyar hai, par live server par Anthropic API key abhi set nahi hai, isliye abhi fixed safe jawab aata hai. Key daalte hi Claude chalu ho jata hai. Red-flag wala emergency reply bina key ke bhi kaam karta hai.
"JWT / bcrypt?"	core/security.py mein JWT/bcrypt ka purana code hai, par live auth portal/auth.py mein hai: scrypt + random session tokens.
"SQLite scale karega?"	Ek gaon-block ke prototype ke liye theek hai. District level par PostgreSQL + PostGIS (+ TimescaleDB for vitals) chahiye; yeh next step hai.
"Triage session restart par?"	Chalti hui triage conversation ka state server memory mein rehta hai, to server restart par woh conversation dobara shuru karni padti hai. Final result (encounter, escalation, audit) database mein safe rehta hai.
"Accuracy / metrics (O1-O6)?"	Repo mein synthetic cohort aur synthetic outbreak simulators aur 4 test suites hain (backend/tests), jo sab pass hote hain. Par clinician panel se 200+ scripted cases ki review (kappa >= 0.7) abhi **nahi hui**; use "next step" bolein, achievement nahi.
"Real data hai?"	Nahi. Demo/seed data saaf kiya gaya hai; alerts sirf asli reports se bante hain. Slides ke screenshots ek alag test database ke hain.

6. Objectives (O1-O6) ke saamne hamara status

#	Objective	Hamara status
O1	Safe voice triage, danger signs 100% escalate, 0 prescriptions	Done by design: 7 fixed rules model se pehle chalte hain; prescription filter. Chest pain Hindi mein live test kiya: EMERGENCY.
O2	BP/sugar deterioration, visit se >= 7 din pehle flag	Trend + adherence engine + synthetic cohort simulator. Real-world validation baaki.
O3	ASHA decision support, clinician se >= 90% agreement	Checklist + protocol citation + doctor AGREE/OVERRIDE se agreement measure hoga. Panel review baaki.
O4	Outbreak >= 3 din pehle, false alarm <= 1/month/block	Multi-stream fusion + baseline + corroboration + 10 km cluster; synthetic outbreak benchmark. Field test baaki.
O5	Multilingual voice, 3+ languages	5 bhasha (hi, en, bn, mr, te) triage + chat + voice. Formal F1 measurement baaki.
O6	Audit 100%, doctor summary <= 30 s	Hash-chained audit log; summary ~1 second mein.

7. Live demo ka sahi order (2-3 minute)

1. **Patient screen (Hindi):** BP/sugar tiles, "kya aaj dawai li?", ASHA ka "call" button, 108 button, language toggle.
2. **ASHA -> Voice Triage:** mareez chuno, consent tick, Hindi chuno, bolo/type karo "seene mein tez dard hai, baayein haath mein ja raha hai aur pasina aa raha hai". Result: **Emergency** + red-flag rule code + checklist.
3. **Doctor:** same case queue mein, mareez ka naam, "what the patient said", one-page summary, AGREE/OVERRIDE.
4. **Officer -> Signals:** village chart (clinic + chemist + school) aur "scoring ke liye 7 din ka baseline chahiye" wali baat.
5. **Audit Trail:** har step ka hash-chained record.

8. Agla kadam (roadmap)

- Clinician panel se 200+ scripted cases par triage accuracy (kappa) aur red-team testing.
- PostgreSQL + PostGIS + TimescaleDB, aur offline-first sync ke saath district scale.
- Bhashini / IndicConformer jaisa dedicated Indic STT; IVR / missed-call feature phone ke liye.
- Claude API key lagakar patient chat live; protocol RAG (WHO/IMNCI/NHM documents) ke saath.
- Play Store (TWA/Bubblewrap), push notifications, password reset.

Live prototype: https://16-4-93-27.sslip.io | Team: Ankit Pandey, Vivek Prasad, Anshika Varshney, Unnati Khandelwal